

11/20 Meeting

Presenter : Yu-Chi Chung

Progress

10/16 ~ 11/06

Per-Scene 3DGS Open-vocab Segmentation



LangSplat (CVPR 2024)
Feature 3DGS (CVPR 2024)
LEGaussians (CVPR 2024)
Gaussian Grouping (ECCV 2024)
Dr. Splat (CVPR 2025)
3DVLGaussian (ICLR 2025)

11/07 ~ 11/20

Feed-Forward 3DGS Open-vocab Segmentation



1. **3DGS** → backbone → **3D scene field with open-vocabulary semantics**

e.g. SceneSplat (ICCV 2025)

2. **Sparse Images** → backbone → **3D scene field with open-vocabulary semantics**

e.g. SLGaussian (ACM MM 2025), SemanticSplat (arXiv 2025), GSemSplat (arXiv 2024)
Uni3R (arXiv 2025), UniForward (arXiv 2025), SpatialSplat (ICCV 2025)

Existing **Feed-Forward** Methods for 3DGS Open-Vocab Segmentation

1. **3DGS** → backbone → 3D scene field with open-vocabulary semantics

- More **precise** Gaussians, better open-vocabulary segmentation.
- Evolved from **per-scene** 3DGS open-vocab segmentation.

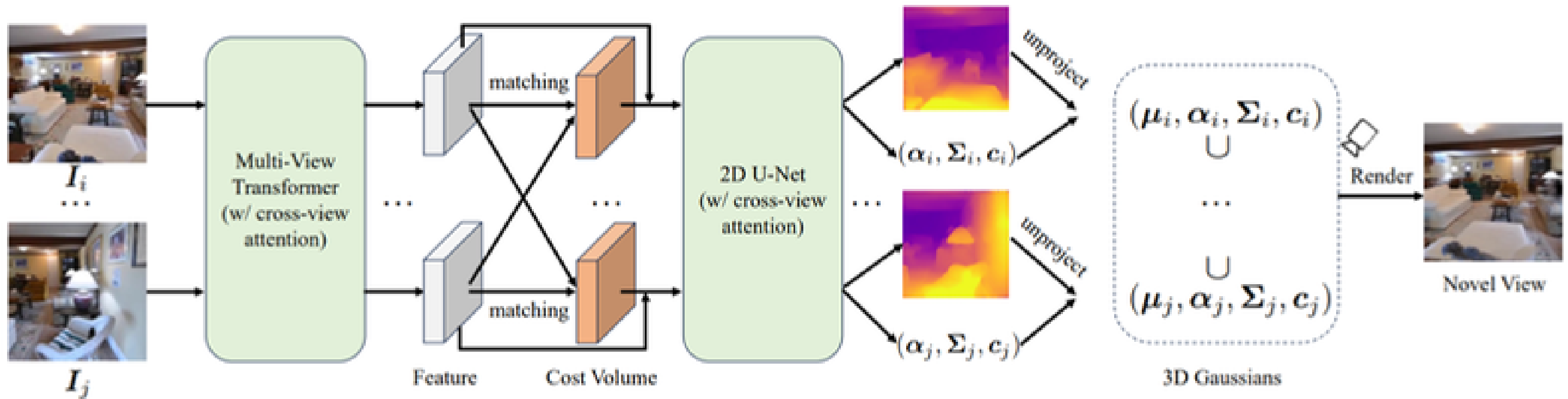
2. **Sparse Images** → backbone → 3D scene field with open-vocabulary semantics

- **Sparse views or pose-free**, enabling usable open-vocabulary segmentation.
- Evolved from **generalizable 3D Gaussian Splatting architectures**.

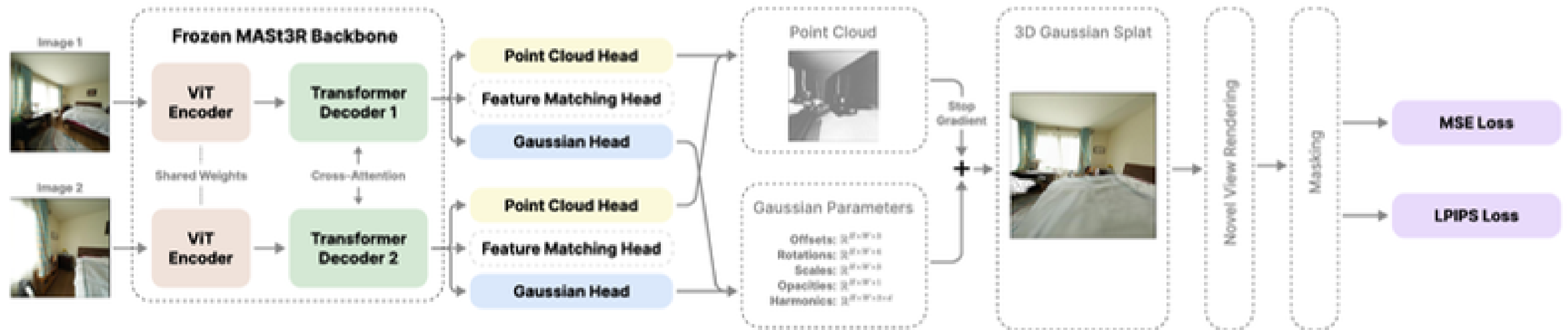
Sparse Images → backbone → 3D scene field e.g. MVSplat, Splatt3R

Generalizable 3D Gaussian Splatting Architectures

1. MVSPLAT: EFFICIENT 3D GAUSSIAN SPLATTING FROM SPARSE MULTI-VIEW IMAGES (ECCV 2024)

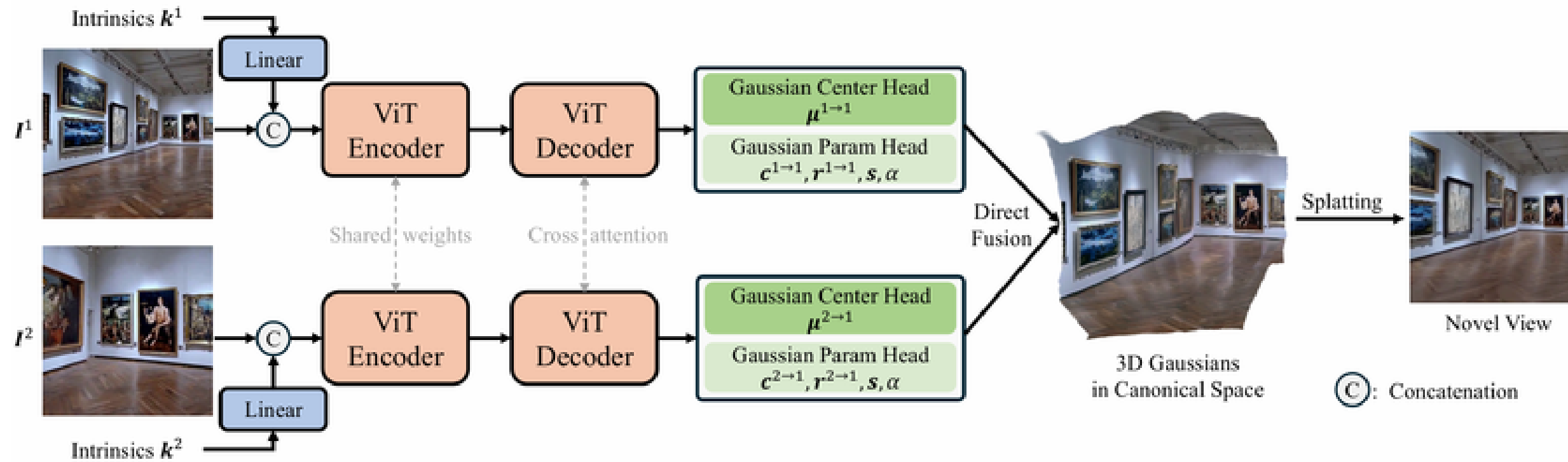


2. SPLATT3R: ZERO-SHOT GAUSSIAN SPLATTING FROM UNCALIBRATED IMAGE PAIRS (ARXIV 2024)

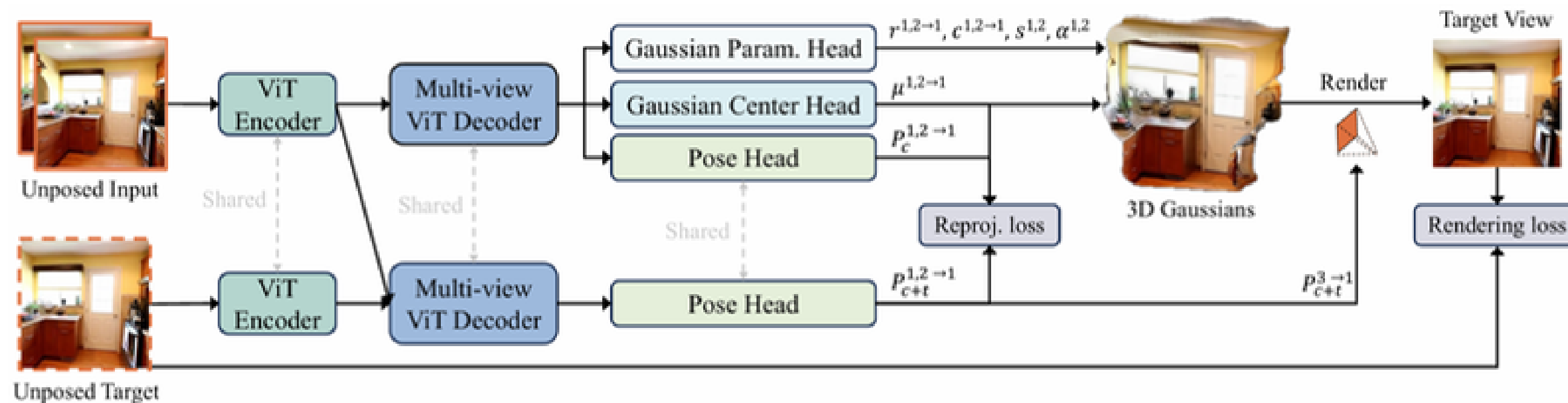


Generalizable 3D Gaussian Splatting Architectures

3. NO POSE, NO PROBLEM: SURPRISINGLY SIMPLE 3D GAUSSIAN SPLATS FROM SPARSE UNPOSED IMAGES (ICLR 2025 ORAL)



4. NO POSE AT ALL: SELF-SUPERVISED POSE-FREE 3D GAUSSIAN SPLATTING FROM SPARSE VIEWS (ICCV 2025)



Discussion

1. 3DGS → backbone → 3D scene field with open-vocabulary semantics

- More **precise** Gaussians, better open-vocabulary segmentation.
- Evolved from **per-scene** 3DGS open-vocab segmentation.
- **Pro: Less competition; high potential for novelty.**
- **Con: Non-intuitive pipeline (requires offline 3DGS generation).**

2. Sparse Images → backbone → 3D scene field with open-vocabulary semantics

- **Sparse views or pose-free**, enabling usable open-vocabulary segmentation.
- Evolved from **generalizable 3D Gaussian Splatting architectures**.
- **Pro: Practical & Scalable; built on fast-evolving backbones.**
- **Con: Crowded research space ?**



SemanticSplat: Feed-Forward 3D Scene Understanding with Language-Aware Gaussian Fields

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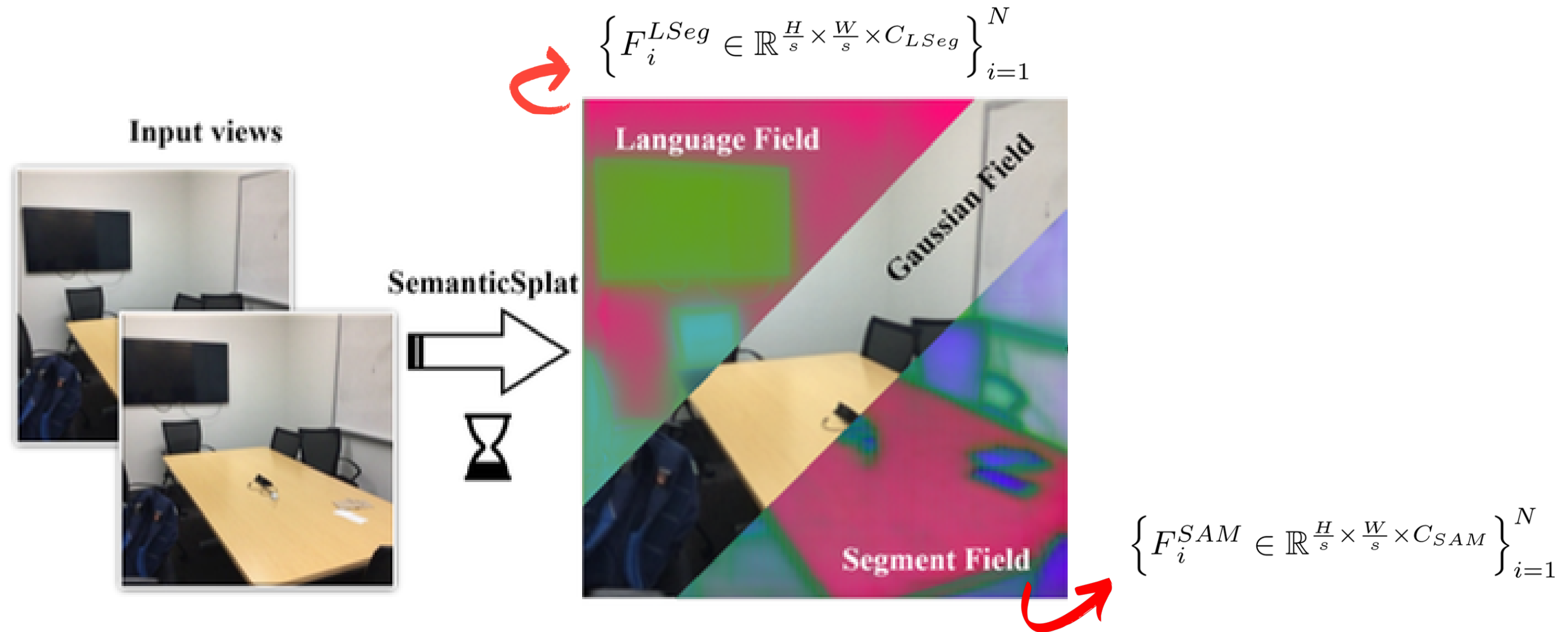
Tsinghua University

Beijing Normal University

arXiv 2025 06

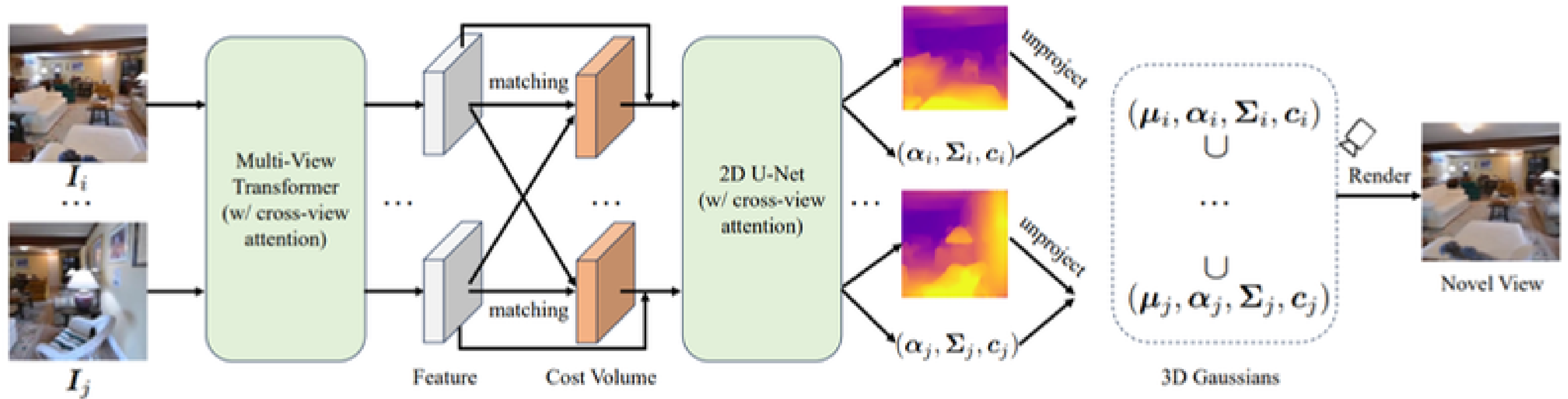
Motivation & Introduction

Per-scene optimization methods **rely on dense input views**, which reduces practicality.



sparse view images → **backbone** → **3D semantic Gaussian field**

Related Work – MVSplat



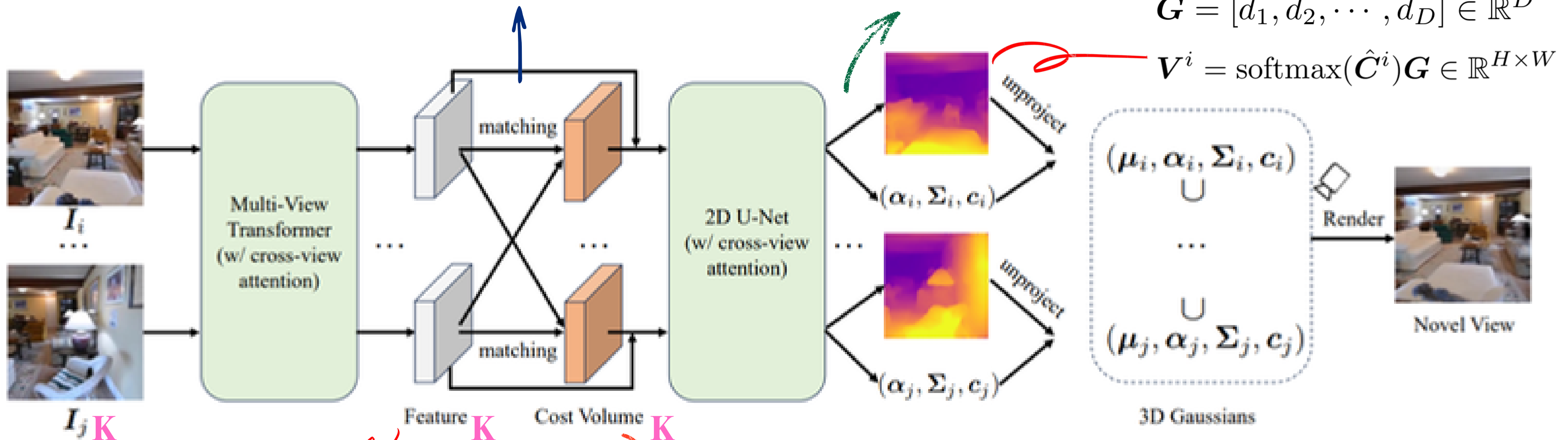
Related Work – MVSpLat

$$\mathbf{F}_{d_m}^{j \rightarrow i} = \mathcal{W}(\mathbf{F}^j, \mathbf{P}^i, \mathbf{P}^j, d_m) \in \mathbb{R}^{\frac{H}{4} \times \frac{W}{4} \times C}, \quad m = 1, 2, \dots, D,$$

$$\tilde{C}^i = C^i + \boxed{\Delta C^i} \in \mathbb{R}^{\frac{H}{4} \times \frac{W}{4} \times D} \quad \hat{C}^i \in \mathbb{R}^{H \times W \times D}$$

$$\mathbf{G} = [d_1, d_2, \dots, d_D] \in \mathbb{R}^D$$

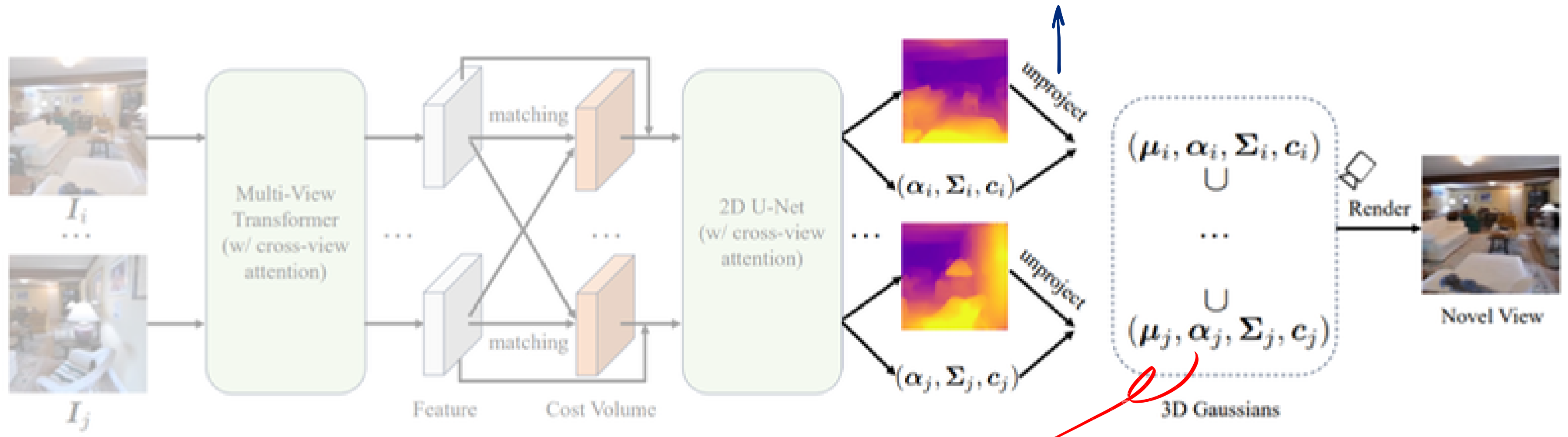
$$\mathbf{V}^i = \text{softmax}(\hat{\mathbf{C}}^i) \mathbf{G} \in \mathbb{R}^{H \times W}$$



$$\{\mathbf{F}^i\}_{i=1}^K \quad (\mathbf{F}^i \in \mathbb{R}^{\frac{H}{4} \times \frac{W}{4} \times C}) \quad \mathbf{C}^i = [\mathbf{C}_{d_1}^i, \mathbf{C}_{d_2}^i, \dots, \mathbf{C}_{d_D}^i] \in \mathbb{R}^{\frac{H}{4} \times \frac{W}{4} \times D} \quad \mathbf{C}_{d_m}^i = \frac{\mathbf{F}^i \cdot \mathbf{F}_{d_m}^{j \rightarrow i}}{\sqrt{C}} \in \mathbb{R}^{\frac{H}{4} \times \frac{W}{4}}, \quad m = 1, 2, \dots, D$$

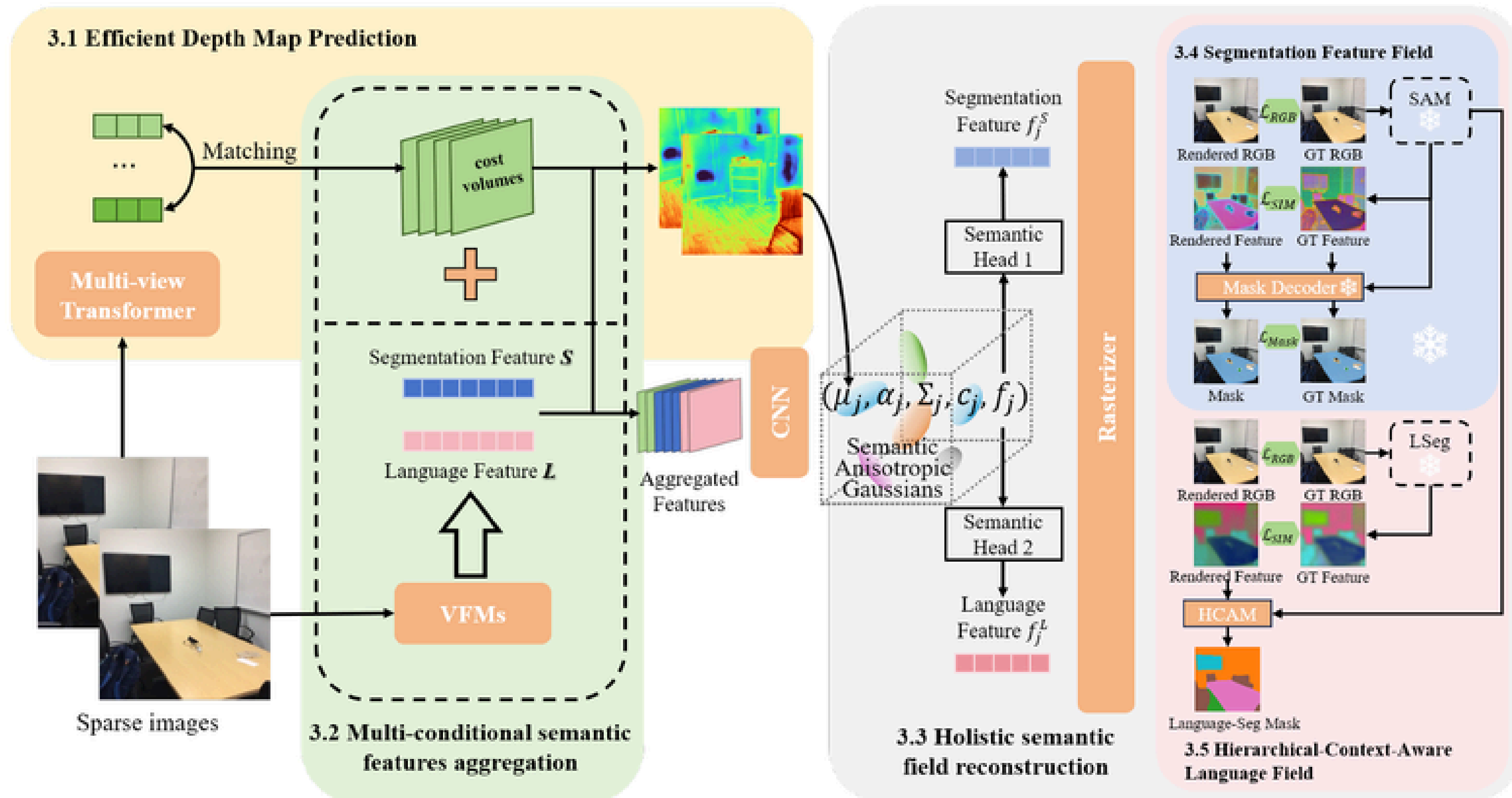
Related Work – MVSplat

$$\mathcal{I} = \{\mathbf{I}^i\}_{i=1}^K, (\mathbf{I}^i \in \mathbb{R}^{H \times W \times 3}) \quad \mathcal{P} = \{\mathbf{P}^i\}_{i=1}^K, \mathbf{P}^i = \mathbf{K}^i[\mathbf{R}^i | \mathbf{t}^i] \quad \mathbf{V}^i = \text{softmax}(\hat{\mathbf{C}}^i) \mathbf{G} \in \mathbb{R}^{H \times W}$$

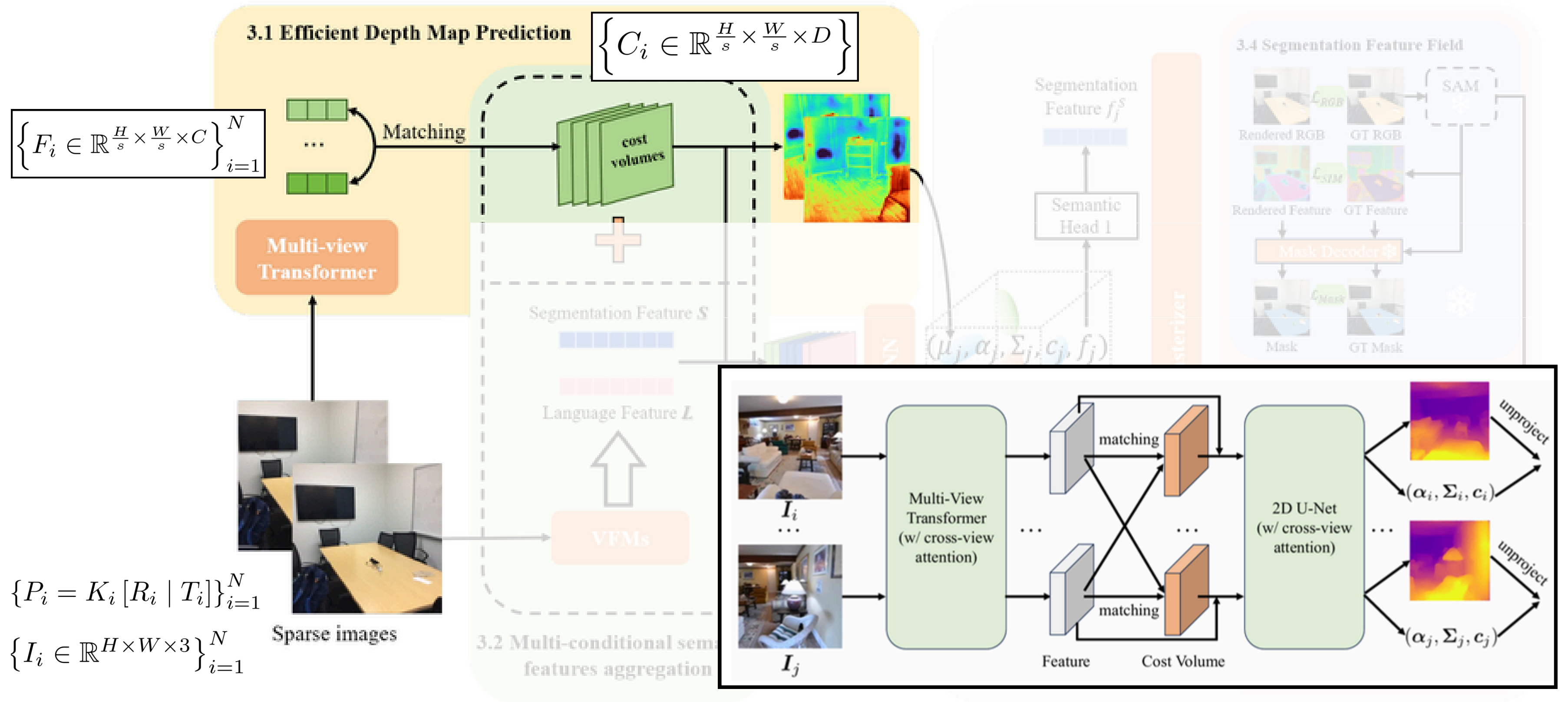


$$\mathbf{C}^i = [\mathbf{C}_{d_1}^i, \mathbf{C}_{d_2}^i, \dots, \mathbf{C}_{d_D}^i] \in \mathbb{R}^{\frac{H}{4} \times \frac{W}{4} \times D} \quad \text{softmax}(\hat{\mathbf{C}}^i)$$

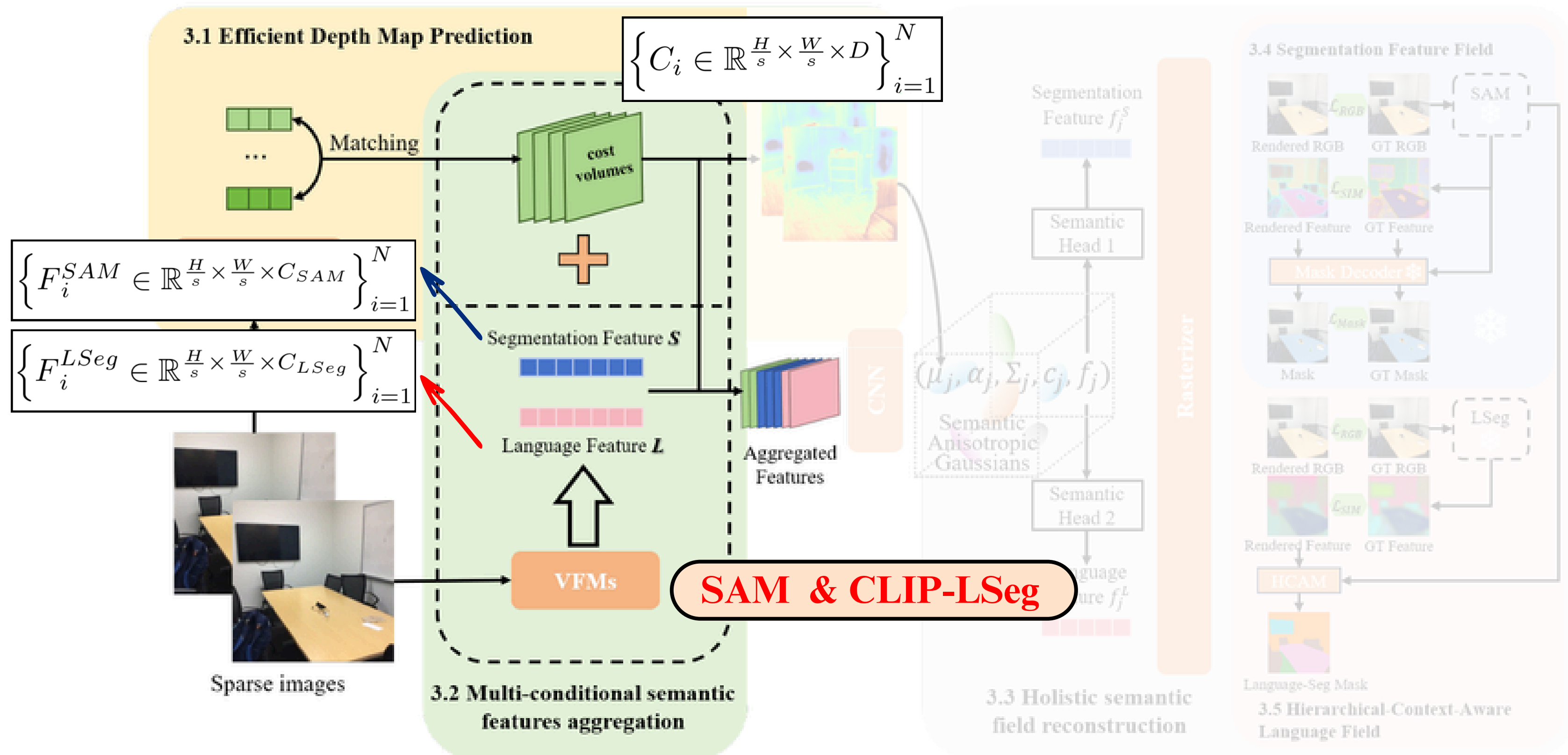
Overview



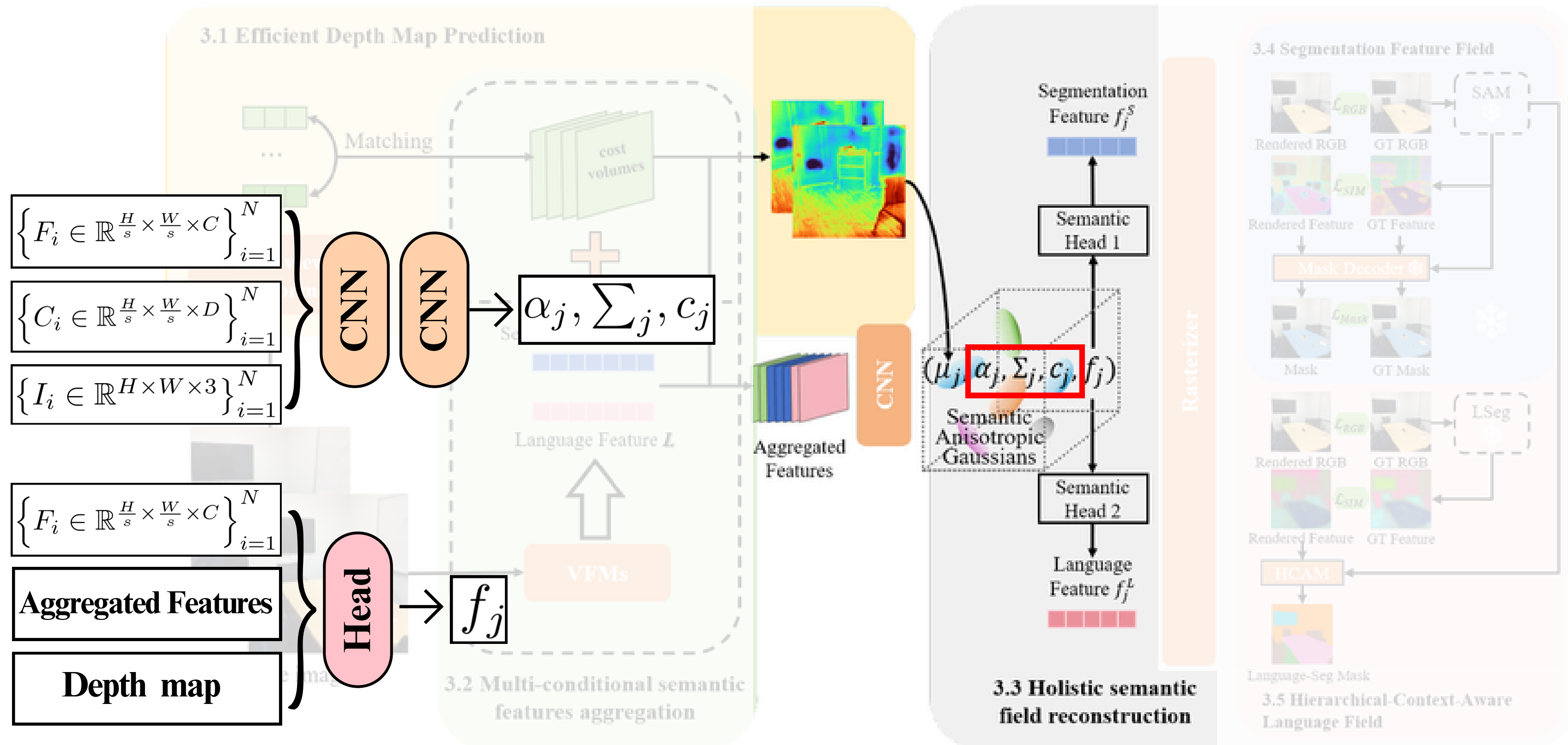
3.1 Efficient Depth Map Prediction



3.2 Multi-conditioned Semantic Features Aggregation



3.3 Semantic Anisotropic Gaussians Prediction



3.4. Segmentation Feature Field

$$\mathcal{L}_{\text{rgb}} = \sum_i \|\mathcal{R}_c(\mu, \alpha, \Sigma, c, f)_i - I_i^{\text{gt}}\|_1 + \lambda_1 \cdot \text{LPIPS}(\mathcal{R}_i, I_i^{\text{gt}})$$

$$L_{\text{dist}}^{\text{Seg}} = 1 - \mathbf{sim}(f_{\text{expand}}(S), \hat{S}) = 1 - \frac{S \cdot \hat{S}}{\|S\| \cdot \|\hat{S}\|}$$

$$S = \left\{ S_i \in \mathbb{R}^{h' \times w' \times d'} \right\}_{i=1}^N \quad \hat{S} = \left\{ \hat{S}_i \in \mathbb{R}^{H' \times W' \times C'} \right\}_{i=1}^N$$

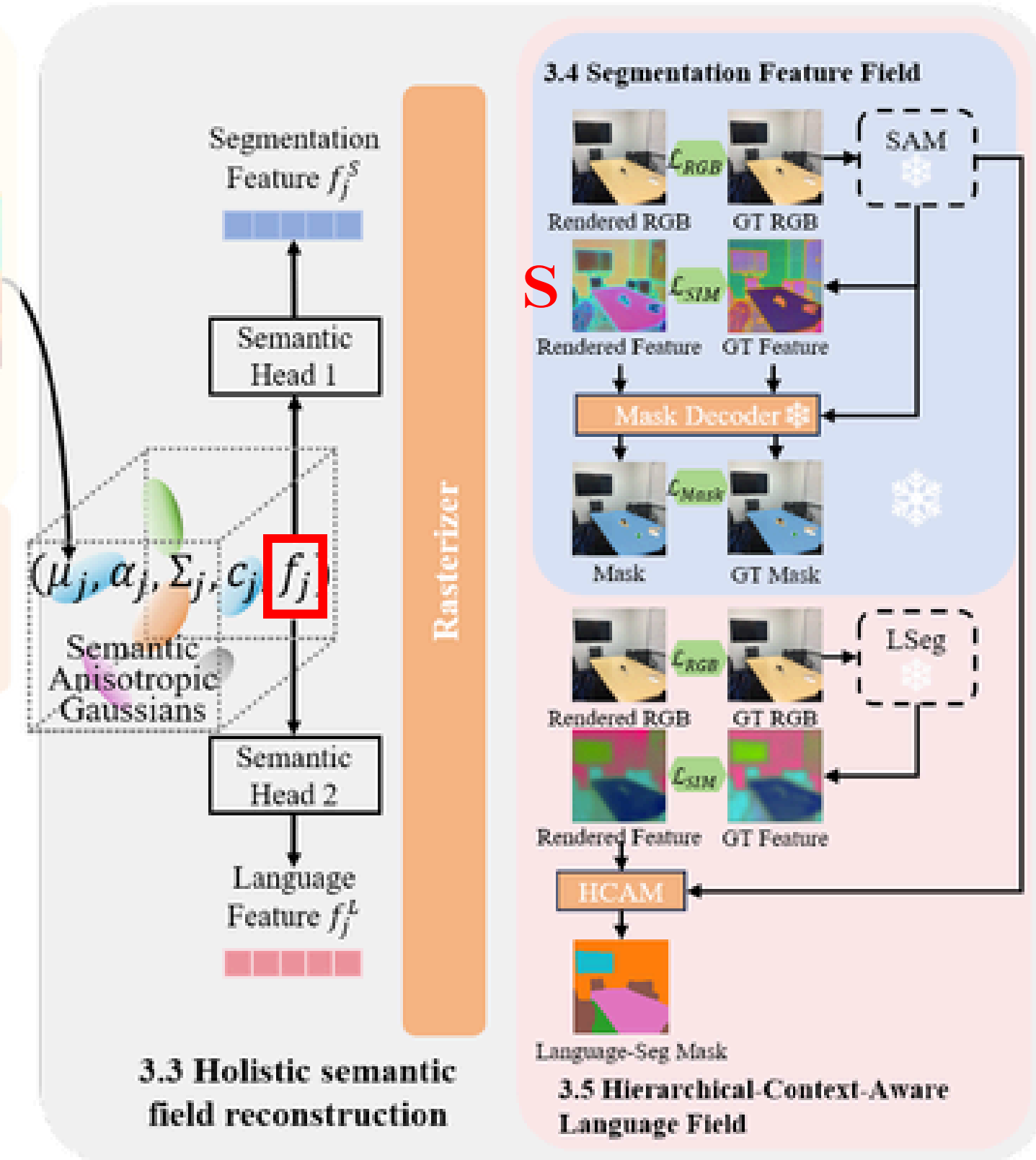
Focal Loss:

$$FL(p_t) = -\alpha(1 - p_t)^\gamma \log(p_t)$$

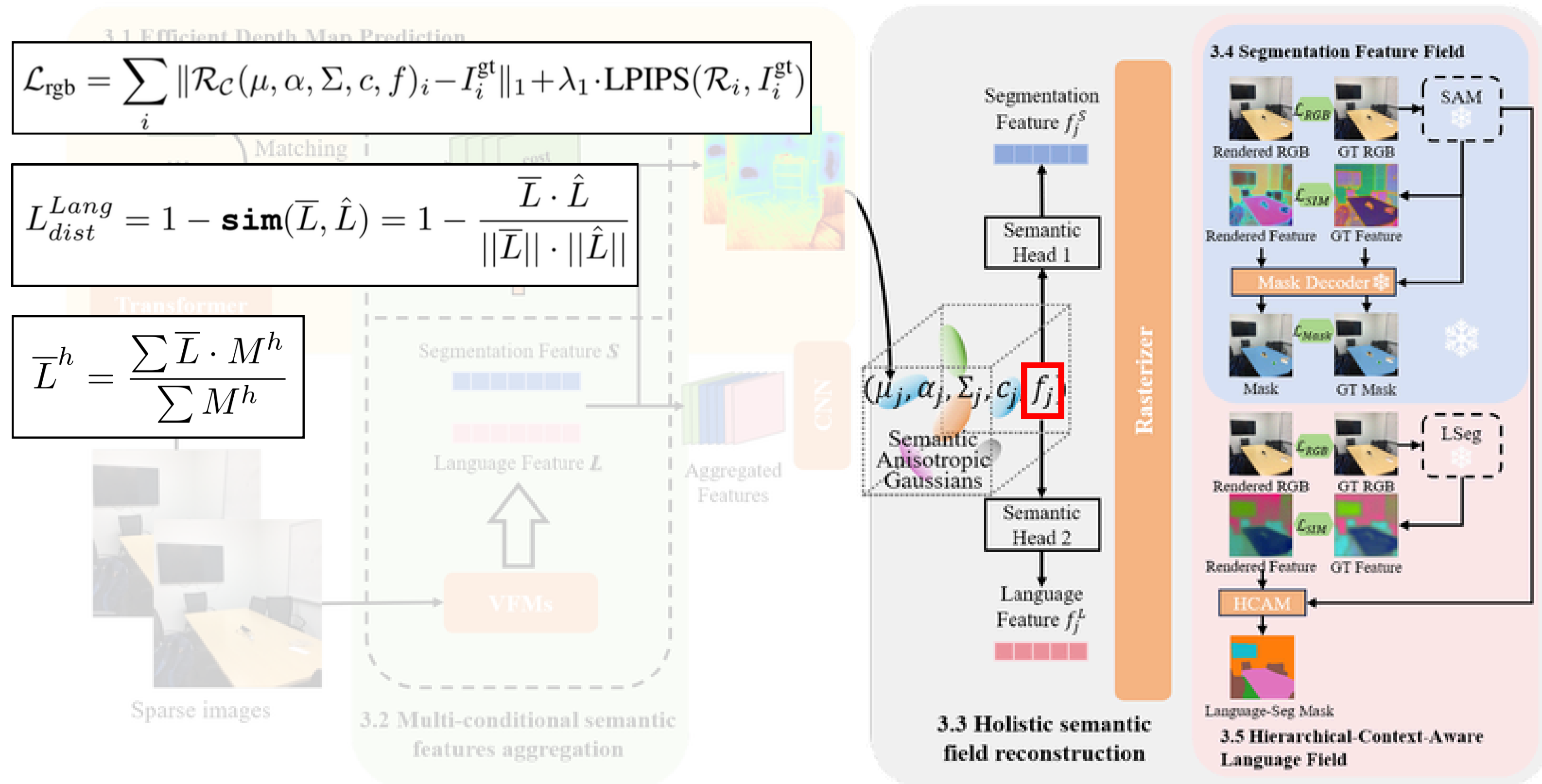
Dice Loss:

$$d = 1 - \frac{2|X \cap Y|}{|X| + |Y|}$$

$$L_{\text{mask}}^{\text{Seg}} = FL(p_t) + \frac{1}{20}d$$



3.5. Hierarchical-Context-Aware Language Field

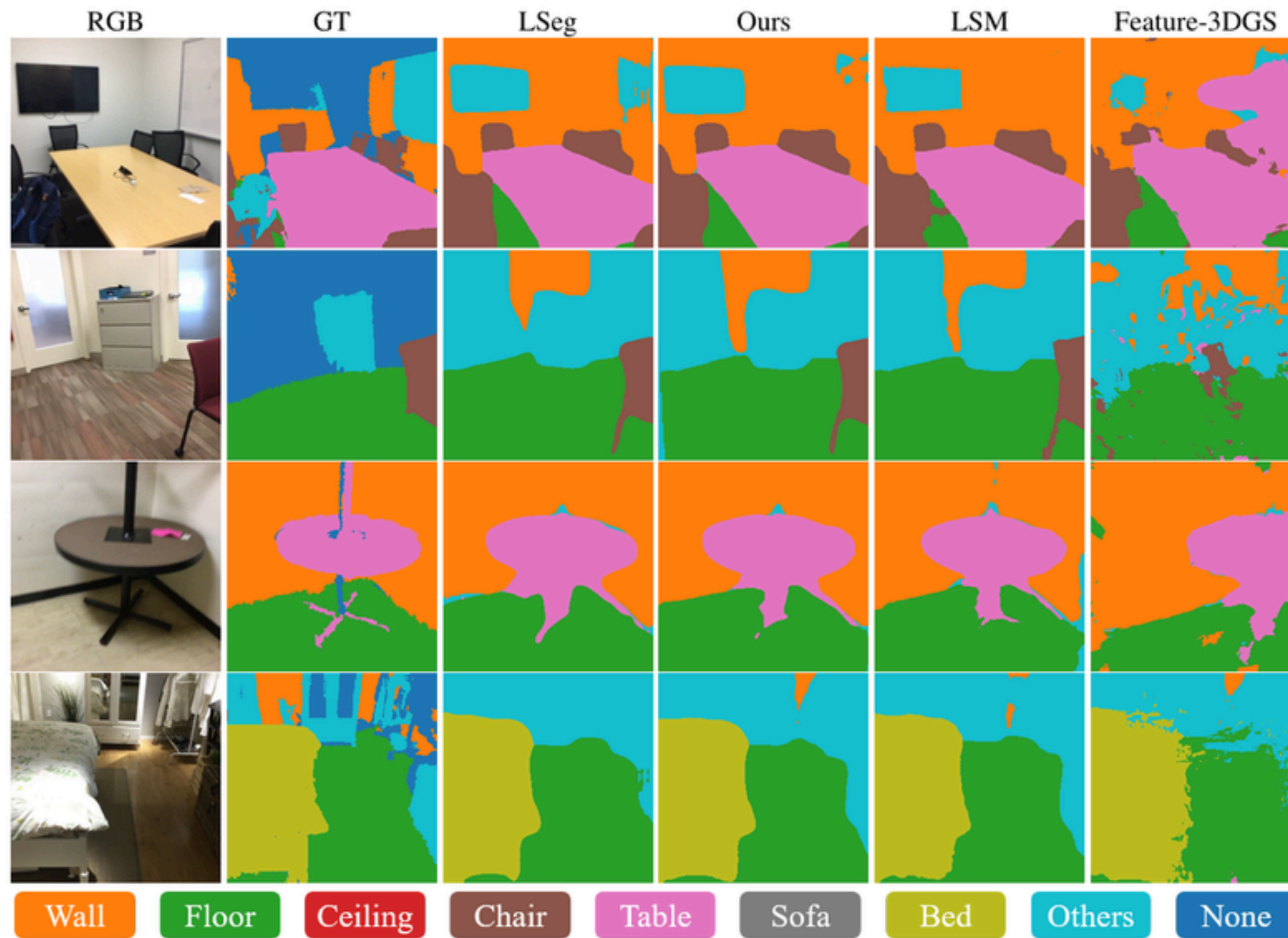


Experiments

Table 1. **Comparison (Language-Seg)**. Performance metrics for source and target view segmentation across different methods.

Method	Source View		Target View				
	mIoU↑	Acc.↑	mIoU↑	Acc.↑	PSNR↑	SSIM↑	LPIPS↓
MVSplat	-	-	-	-	23.87	0.820	0.201
LSeg	0.365	0.694	0.364	0.693	-	-	-
LSM	0.347	0.679	0.347	0.679	17.84	0.630	0.372
Feature-3DGS	0.510	0.804	0.235	0.585	13.00	0.407	0.600
Ours	0.376	0.707	0.371	0.702	21.88	0.879	0.191
Ours w/HCAM	0.376	0.707	0.386	0.710	21.88	0.879	0.191

Experiments



Thanks for your attention!